Structural Ambiguity Amplification in Near-Critical Hawkes Market Making

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Abstract

Robust market-making models usually perturb drift, volatility, or fill rates. We study ambiguity in the self-excitation structure of order flow. In a Hawkes-driven limit order book, this structure is encoded by a branching matrix Γ, whose spectral radius $\rho(\Gamma)$ controls proximity to criticality. We formulate a reduced robust inventory market-making model under structural uncertainty over Γ and show, at the risk-premium/proxy level, that the ambiguity premium is governed by the Hawkes resolvent $(I - \Gamma)^{-1}$. A key correction is that the critical exponent depends on the ambiguity normalization: relative critical-slack ambiguity gives a square law, while absolute branching-matrix ambiguity adds one derivative factor. A reproducible experiment package validates the scaling distinction, evaluates robust and nominal quoting policies with common random numbers, solves a finite-scenario robust Bellman approximation, validates Hawkes simulation, and checks public LOBSTER equity, Binance aggregate-trade, and crypto order-book samples for clustering, liquidity overdispersion, and calibration fragility. A focused event-queue backtest adds a reduced execution check for quote/no-quote behavior on Ogata event times, and level-10 LOBSTER replays add displayed-depth, residual-volume priority, sensitivity, and observable reconstruction diagnostics. The appendix states a compact continuous-intensity HJBI verification theorem for the truncated multitype Hawkes control game under the standard compact-PDMP dynamic-programming and comparison hypotheses. An untruncated Lyapunov theorem supplies nonexplosion, $p > 1$ weighted moment bounds, and truncation-tail control under a common robust subcriticality condition. A weighted comparison theorem gives uniqueness of the untruncated viscosity solution in weighted subclasses with a fixed boundary condition at infinity.

Keywords. Hawkes processes; robust market making; structural ambiguity; near-critical order flow; viscosity solutions; limit order books.

MSC. Primary 91G80, 60G55; Secondary 93E20, 49L25, 60J76.

1 Introduction

Liquidity withdrawal during stress is often modeled as a response to volatility or inventory risk. MESA studies a sharper mechanism: if order flow is self-exciting and near critical, then uncertainty about the excitation structure itself becomes expensive. The same estimation error in Γ is benign far from criticality and severe when $\rho(\Gamma)$ approaches one.

The intended target for this version is *SIAM Journal on Financial Mathematics*. The contribution is deliberately narrower than the original MESA framework draft. Hawkes market making, robust market making, near-critical Hawkes limits, and graphon mean-field games all have substantial prior art. The publishable intersection is structural Hawkes ambiguity for market making, the resulting spectral amplification law, and empirical diagnostics showing why naive near-critical Hawkes calibration must be stress-tested before it is used inside a market-making simulator.

2 Related Work And SOTA Positioning

Table 1 summarizes the current positioning. Hawkes market making has been studied in point-process control [2], neural Hawkes LOB simulation [7, 3], adversarial/Hawkes RL [6], impulse control [10], attention-based LOB-feature DRL [8], and latency-aware RL [9]. Robust quote refusal is also active [5]. Recent multivariate LOB forecasting, nearly unstable Hawkes limits, and simulator/state-dependent Hawkes papers emphasize realism, calibration, and local instability [4, 14, 15, 11, 12, 13]. Financial criticality and price-discovery mechanisms provide additional motivation for treating near-instability carefully [16, 17]. MESA does not claim to replace these simulators or RL agents. Its distinct role is to stress-test the geometry of branching-matrix uncertainty itself.

The reproducibility package records source URLs, code URLs when visible, and code-reality notes for this comparison. The 16 May 2026 web audit found the strongest external code baseline in El Karmi's deterministic Hawkes LOB simulator; accordingly, MESA does not claim simulator primacy and instead positions its contribution as a spectral ambiguity-risk layer.

3 Model

Let $N_t^{k,s}$ denote marked order arrivals by type k and side s. The exponential Hawkes intensity is

$$\lambda_t^{k,s} = \mu^{k,s} + \sum_{j,\sigma} \int_0^{t^-} \alpha_{j,\sigma}^{k,s} \beta e^{-\beta(t-u)} \, dN_u^{j,\sigma}.$$

The branching matrix Γ has entries equal to integrated excitation. The process is stationary when $\rho(\Gamma) < 1$, and

$$\bar{\lambda} = (I - \Gamma)^{-1} \mu.$$

A market maker chooses bid and ask offsets $u_t = (\delta_t^b, \delta_t^a) \in \mathcal{U}$ with inventory $q_t \in \{-Q, \dots, Q\}$. Nature chooses a structural model from an ambiguity set

$$\mathcal{A}_\epsilon = \{\Gamma \geq 0 : \|\Gamma - \widehat{\Gamma}\|_F \leq \epsilon, \ \rho(\Gamma) \leq \rho_{\max} < 1\}.$$

In the computational version used here, Nature is represented by a finite set containing the nominal matrix and Perron-aligned boundary perturbations.

4 Theory Repair

Assumption 1 (Perron regularity)**.** *Γ is nonnegative and irreducible. Its Perron root $\rho < 1$ is simple, the spectral gap is bounded below, and left/right Perron vectors ℓ, r are normalized by $\ell^\top r = 1$ with bounded condition number.*

Work	Hawkes MM	Robust MM	Γ uncertainty	Critical exponent	MESA position
Law–Viens 2019/2020	no	no	no	no	Marked point-process/HJB-QVI LOB MM benchmark; MESA adds Hawkes branching-matrix ambiguity.
Wang–Ventre–Polukarov 2023/2025 quote/no-quote	no	yes	no	no	Robust quote-refusal/single-sided quoting benchmark; Hawkes branching uncertainty is outside its scope.
Wang–Ventre–Polukarov 2024/2025 Hawkes ARL	yes	yes	no	no	Closest Hawkes/ARL policy benchmark; no branching-matrix amplification law.
Jain et al. 2025 Hawkes impulse control	yes	no	no	no	Closest HJB-QVI benchmark; robustness to Γ is not the main object.
Lalor–Swishchuk 2025 neural Hawkes LOB	yes	no	no	no	Neural Hawkes LOB simulator/DRL market-making baseline; ambiguity amplification is complementary.
Kumar 2021/2024 Deep Hawkes	yes	no	no	no	Deep Hawkes high-frequency MM/simulation benchmark; MESA adds structural Γ ambiguity diagnostics.
Guo–Lin–Huang 2023 Attn-LOB DRL	no	no	no	no	LOB-feature DRL baseline; focus is alpha learning rather than Hawkes structure.
Jiang et al. 2025 latency-aware RL	no	no	no	no	Latency/inventory-aware neural MM benchmark; robust Hawkes ambiguity is not modeled.
Raffaelli et al. 2026 multivariate Hawkes BTC LOB	no	no	no	no	Current multivariate LOB calibration/forecasting benchmark; robust control is outside its scope.
El Karmi 2025 deterministic LOB simulator	no	no	no	no	Simulator-realism baseline with Hawkes-driven order flow; robustness is complementary.
Noble–Rosenbaum–Souilmi 2026 realism gap	no	no	no	no	Strong simulator-realism and execution-sensitivity benchmark; MESA is a complementary risk layer.
Kimura 2026 state-dependent Hawkes	no	no	no	no	State-dependent Hawkes LOB with local supercriticality evidence; MESA targets robust Γ ambiguity.
Szymanski–Xu 2025 nearly unstable Hawkes	no	no	no	yes	Near-critical Hawkes scaling limits; MESA adds market-making control and Γ-ambiguity premiums.
El Karmi 2026 bivariate nearly unstable Hawkes	no	no	no	yes	Current bivariate near-critical Hawkes limit theory; MESA targets robust market-making premiums.
MESA reduced-form package	yes	yes	yes	yes	Reduced-form theorem with policy stress tests and calibration diagnostics.

Table 1: Comparison with nearby work. The point is not to claim simulator primacy, but to isolate the structural Hawkes ambiguity layer.

Proposition 1 (Resolvent amplification)**.** *Under Perron regularity,*

$$(I - \Gamma)^{-1} = \frac{r\ell^\top}{1 - \rho(\Gamma)} + O(1/\text{gap}).$$

If the price-risk loading has nonzero projection on the Perron mode, the effective order-flow risk inherits this singularity.

Proposition 2 (Perturbation direction)**.** *For a small perturbation E,*

$$\rho(\Gamma + E) - \rho(\Gamma) = \ell^\top E r + O(\|E\|_F^2/\text{gap}).$$

The first-order Frobenius direction that maximally increases the Perron root is the rank-one Perron direction.

Theorem 1 (Reduced-form normalization-dependent ambiguity premium)**.** *Suppose the robust half-spread is locally linear in an effective Hawkes variance proxy proportional to $(1 - \rho)^{-2}$. Then relative critical-slack ambiguity, $\Delta\rho = O(\epsilon_{\text{rel}}(1 - \rho))$, gives*

$$\psi = O\left(\frac{\epsilon_{\text{rel}}}{(1 - \rho)^2}\right).$$

Absolute branching-matrix ambiguity, $\Delta\rho = O(\epsilon)$, gives the more singular derivative law

$$\psi = O\left(\frac{\epsilon}{(1 - \rho)^3}\right).$$

Matching lower bounds require Perron alignment and an active worst-case perturbation.

Remark 1. *The theorem is intentionally stated as a reduced-form risk-premium result. The companion proof appendix gives scalar and finite-dimensional Perron-visible reduced-form versions of this bridge, a finite-state Bellman verification theorem, and a compact continuous-intensity HJBI verification theorem for a truncated multitype Hawkes control game, plus an untruncated Lyapunov stability bridge, pairwise weighted comparison theorem, and local differentiability theorem for smooth interior robust quote optimizers. What remains open beyond this model class is no-quote/boundary active-set differentiability, removal of the compact-PDMP DPP/comparison hypotheses, and extension to state-dependent, power-law, or learned Hawkes kernels.*

5 Numerical Method

The experiment package implements three levels.

1. Scaling diagnostics for scalar and multitype resolvent proxies.
2. Common-random-number policy evaluation under full simulated Hawkes paths.
3. A finite-scenario robust inventory Bellman solver for a scalar reduced state with side-level quote/no-quote actions.
4. A reduced event-queue backtest on Ogata event times with queue-ahead mechanics and no-quote side-time accounting.
5. Displayed-depth, residual-volume priority-aware level-10, and deepest-public level-50 LOBSTER replays on public message/order-book streams.
6. Observable level-10 order-book reconstruction from messages, checked against official LOBSTER snapshots.

7. Ogata-thinning validation and time-step bias diagnostics.
8. LOBSTER Hawkes diagnostics over event type, time scale, marks, and side labels.
9. Binance aggregate-trade event diagnostics for public crypto clustering.

The Bellman solver evaluates a side-level action set in which each side may either quote at an offset or withdraw:

$$\begin{aligned}h_n(q) = \max_{a^b,\delta^b,a^a,\delta^a} \min_{\Gamma\in S} \Big\{ & -\Delta t \left(\phi q^2 + \frac{\gamma}{2}\sigma^2(\Gamma)q^2\right) \\ & + p_b(\delta^b + h_{n+1}(q+1)) + p_a(\delta^a + h_{n+1}(q-1)) \\ & + (1-p_b-p_a)h_{n+1}(q)\Big\}.\end{aligned}$$

Here $a^b, a^a \in \{0,1\}$ are side activity indicators and p_b, p_a are set to zero on inactive sides.

Theorem 2 (Finite-state Bellman verification)**.** *Consider a finite inventory grid, finite Hawkes-regime grid, finite quote/no-quote action set, bounded rewards, bounded terminal payoff, and a finite rectangular ambiguity set S of stable branching matrices. The robust finite-regime market-making game has a value. Its value is the unique solution of the backward Bellman recursion, equivalently the finite-state HJBI ODE*

$$-\partial_t V(t,x) = \max_{u\in U(x)} \min_{\Gamma\in S} \left\{ r(x,u,\Gamma) + \sum_y L^{u,\Gamma}(x,y)V(t,y) \right\}, \qquad V(T,x) = g(x),$$

and deterministic Markov selectors attain the finite max–min. Thus the implemented finite-scenario quote/no-quote DP is a verified robust control approximation. The continuous-intensity multitype Hawkes HJBI is outside this finite-state theorem, and is handled by the compact, Lyapunov, and weighted comparison theorems that follow.

Theorem 3 (Compact continuous-intensity HJBI verification)**.** *Consider the truncated multitype Hawkes control state $X_t = (q_t, \lambda_t)$, with finite inventory $q_t \in \{-Q,\dots,Q\}$ and compact intensity domain $\lambda_t \in K \subset \mathbb{R}^d_+$. Between jumps, $d\lambda_t = \beta(\mu - \lambda_t)\,dt$, and a mark-$i$ jump maps λ to $\Pi_K(\lambda + \beta\Gamma e_i)$. Suppose the quote action set and the rectangular ambiguity set $\mathcal{A}$ are compact, all rewards, terminal payoffs, and controlled mark intensities are bounded and continuous, the rates and jump maps are uniformly Lipschitz on each inventory slice, and $\sup_{\Gamma\in\mathcal{A}} \rho(\Gamma) < 1$, and the compact PDMP game satisfies the standard dynamic-programming and bounded-comparison principles for the rectangular admissible class. Then the robust market-making game has a bounded uniformly continuous value function, and it is the unique bounded viscosity solution of*

$$-\partial_t V = \sup_u \inf_{\Gamma\in\mathcal{A}} \left\{ r(q,\lambda,u,\Gamma) + \mathcal{L}^{u,\Gamma} V(q,\lambda) \right\}, \qquad V(T,q,\lambda) = g(q,\lambda),$$

where $\mathcal{L}^{u,\Gamma}$ is the Hawkes PDMP generator. If the value is classically differentiable and the Hamiltonian has measurable saddle selectors, those selectors verify the robust value. Thus the compact continuous-intensity control bridge is established for the truncated game under the stated compact-PDMP hypotheses; the global untruncated near-critical extension still requires the stability bridge below and a comparison argument on unbounded intensity domains.

Theorem 4 (Untruncated Hawkes Lyapunov bridge)**.** *For the unprojected Hawkes intensity state $\lambda_t \in \mathbb{R}^d_+$, suppose there exist $v > 0$ and $\kappa > 0$ such that $\Gamma^\top v \le (1-\kappa)v$ for every $\Gamma \in \mathcal{A}$, fix $p > 1$, and suppose controlled mark rates satisfy $0 \le \Lambda_i(q,\lambda,u,\Gamma) \le \lambda_i$. Then $W_v(\lambda) = 1 + v^\top \lambda$ satisfies*

$$\mathcal{L}^{u,\Gamma} W_v(q,\lambda) \le C_0 - C_1 W_v(\lambda)$$

uniformly over controls and Nature. Hence the untruncated controlled Hawkes PDMP is non-explosive, has finite pth weighted maximal moments on finite horizons, and compact truncations $\{W_v \le R\}$ approximate the untruncated value with exit probability at most $C_{T,p} W_v(\lambda)^p / R^p$ and linearly-growing payoff tail error of order R^{1-p}. This removes the compact projection for stability and tightness; the next theorem adds a sufficient weighted comparison condition for the unbounded HJBI.

Theorem 5 (Weighted comparison for the untruncated HJBI)**.** *Assume common Lyapunov subcriticality, compact controls and ambiguity, continuous locally Lipschitz controlled rates with $0 \le \Lambda_i \le \lambda_i$, Lipschitz affine Hawkes jump maps, linearly growing rewards, and the weighted boundary condition*

$$\limsup_{R\to\infty} \sup_{W_v(\lambda)\ge R} \frac{U(t,q,\lambda) - V(t,q,\lambda)}{W_v(\lambda)} \le 0$$

for every subsolution U and supersolution V being compared. Then any upper-semicontinuous weighted-growth subsolution of the untruncated HJBI is dominated by any lower-semicontinuous weighted-growth supersolution with larger terminal data. Hence the untruncated robust Hawkes control value is the unique viscosity solution in any W_v-weighted subclass with a fixed boundary condition at infinity, and compact truncation solutions converge locally uniformly to it.

Theorem 6 (Interior robust quote optimizer sensitivity)**.** *Fix a state/adjoint tuple in the HJBI Hamiltonian and parameterize the branching matrix or ambiguity center by θ. Suppose the robust reduced Hamiltonian $\mathfrak{H}(\delta,\theta) = \inf_{\Gamma'\in\mathcal{A}(\theta)} \mathcal{H}(\delta,\Gamma')$ has a unique smooth active worst-case selector, the quote constraints are inactive, and $D^2_{\delta\delta}\mathfrak{H} \preceq -mI$ locally. Then the unique interior quote maximizer $\delta^*(\theta)$ is locally C^1 and*

$$D_\theta \delta^*(\theta_0) = -\left[D^2_{\delta\delta}\mathfrak{H}(\delta_0,\theta_0)\right]^{-1} D^2_{\delta\theta}\mathfrak{H}(\delta_0,\theta_0).$$

Thus an adverse Perron-direction mixed derivative widens the robust half-spread to first order. Boundary no-quote switches, spread caps, or active-adversary changes remain nonsmooth active-set cases.

The implementation is vectorized over quote grids and parallelized over scenarios using up to 16 worker processes on the local 18-core Apple Silicon machine.

6 Experiments

6.1 Scaling Ablations

Object	Family	Mean slope
Relative-slack premium	scalar	-2.000
Absolute-Gamma derivative	scalar	-3.000
Matrix resolvent	rank1	-3.424
Matrix resolvent	block	-3.430
Matrix resolvent	near-degenerate	-3.432
Matrix resolvent	sparse	-3.470

Table 2: Critical exponent ablation.

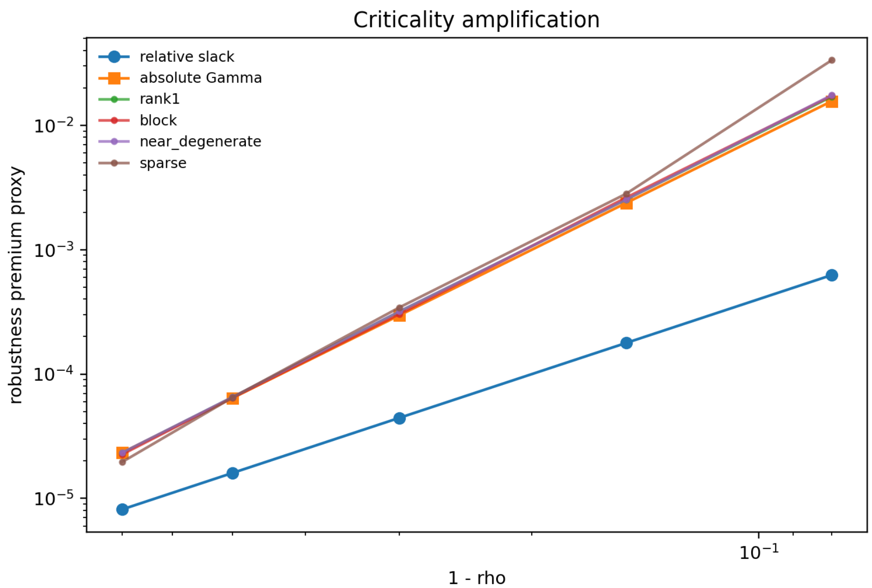

Figure 1: Relative-slack, absolute-Gamma, and matrix-resolvent criticality amplification.

We also run a targeted spectral-gap/visibility ablation using a two-mode branching matrix with eigenvalues ρ and $1-k(1-\rho)$ for $k \in \{2, 4, 8\}$. Perron-visible loadings under a Perron-aligned adversary recover slope -2.00 with median normalized premium ratio 2.8125. A weak Perron-visible loading has the same exponent after visibility normalization, while a Perron-orthogonal loading has no positive Perron-aligned premium. If the adversary instead moves the second eigenmode, the orthogonal loading again has slope -2.00 when the second mode is near critical; for weakly visible loadings the median normalized second-mode coefficient falls from 2.639 at $k = 2$ to 0.0365 at $k = 8$. Thus the lower-bound statement needs both Perron visibility and an ambiguity set that activates the Perron direction; otherwise the leading coefficient can vanish or shift to a different near-critical mode.

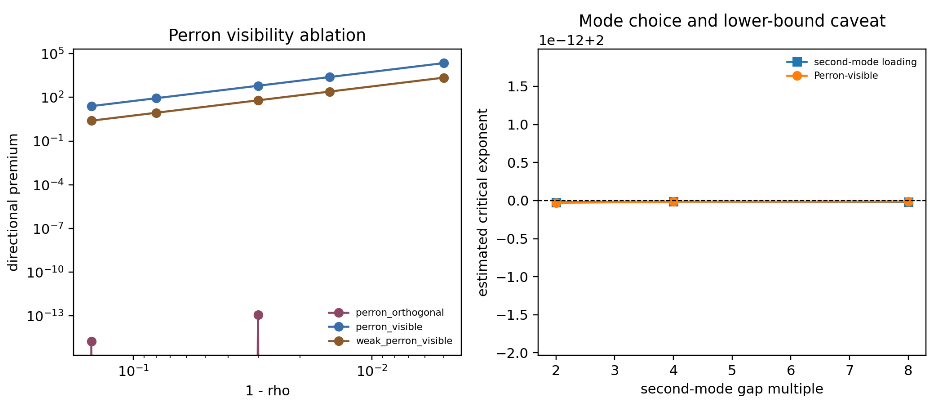

Figure 2: Spectral-gap and Perron-visibility ablation. Perron-visible loadings recover the square law under aligned ambiguity, while orthogonal loadings only respond to second-mode perturbations.

6.2 Policy Ablations

Policy	Certainty equivalent	CVaR5	Mean diff vs nominal
robust_gamma	-13.944	-38.073	0.012
robust_vol_only	-13.956	-38.087	0.000
nominal_hawkes	-13.956	-38.087	baseline
robust_gamma_abs	-13.624	-37.321	0.306
known_true_gamma_no_ambiguity	-13.952	-38.082	0.004
as_poisson	-13.974	-38.113	-0.017
liquidity_guard	-13.624	-37.321	0.306

Table 3: Common-random-number policy ablation.

The absolute-Gamma and liquidity-guard policies deliver the strongest displayed certainty equivalent, CVaR5, and paired mean-wealth differences in the main ablation. The relative-slack robust-Gamma policy is closer to nominal, with a smaller but positive paired mean-wealth difference. We do not interpret the known-true-Γ benchmark as an oracle optimum: it is given the stressed ρ, but it does not carry an ambiguity premium, so it under-hedges model risk in this stress design. The table is best read as evidence that the policy response depends materially on the ambiguity normalization and on the withdrawal rule.

6.3 Policy Time-Step Audit

Because the simulator validation below shows that near-critical Hawkes counts are sensitive to bin width, we reran focused policy comparisons at $\Delta t \in \{0.04, 0.02, 0.01\}$ for the stressed $\epsilon = 0.02$, $\hat{\rho} = 0.92$ scenario. Across the tested time steps, the relative-slack robust-Gamma policy keeps a positive mean-wealth advantage over nominal Hawkes: 0.042–0.060. The absolute-Gamma policy produces much larger mean-wealth gains, 1.154–1.366, but with negative lower paired quantiles, consistent with the broader point that ambiguity normalization changes the economic policy, not only the scale of the premium.

6.4 Ogata Arrival Audit

We also compare the fast binned Hawkes recursion against event times generated by Ogata thinning and then binned onto the same decision grid. In the $\epsilon = 0.02$, $\hat{\rho} = 0.92$ check, the discrete recursion produces more events than the Ogata reference: mean total events are 226.8 under the discrete recursion and 185.5 under Ogata-binned arrivals. The relative-slack robust-Gamma policy is positive relative to nominal under discrete arrivals, with mean-wealth advantage 0.030, but is essentially flat and slightly negative under Ogata-binned arrivals, with mean-wealth difference -0.001. Absolute-Gamma remains strongly positive under both generators, with advantages 0.705 and 0.690. This check weakens any blanket claim that robust-Gamma dominates under every arrival generator, while still showing that the binned-recursion result is not wildly disconnected from an event-time reference.

6.5 Event-Queue Backtest

We then run a reduced event-driven queue backtest on Ogata Hawkes event times. The evaluator reuses each event path across policies, updates quotes every $\Delta t = 0.02$, tracks a simple queue-ahead state, and records side-time spent not quoting. For $\hat{\rho} = 0.92$ over 24 paths, robust-Gamma is nearly

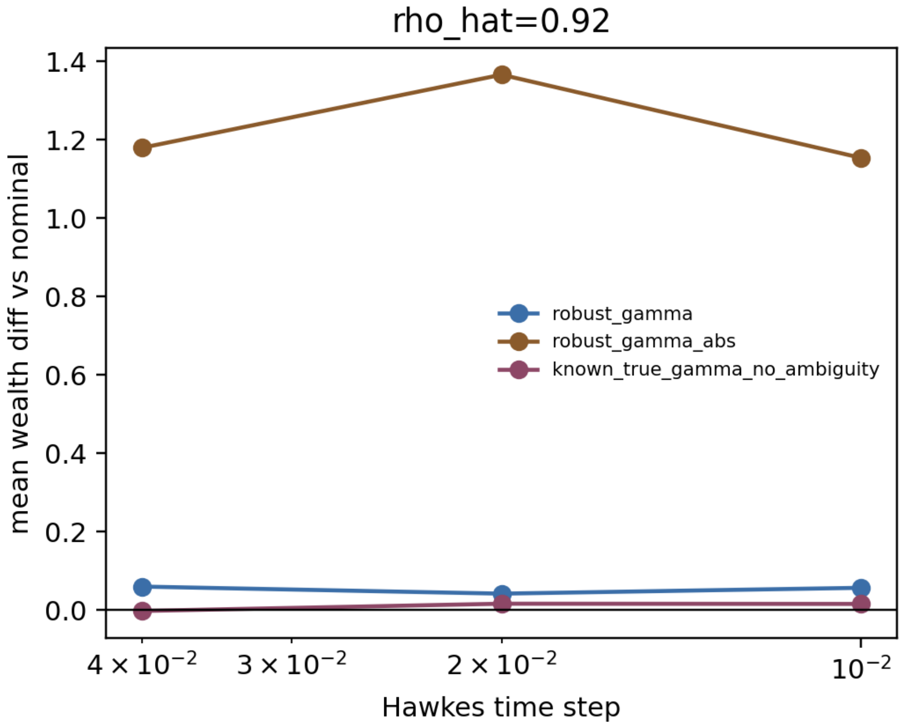

Figure 3: Focused policy time-step audit. Positive values indicate outperformance relative to nominal Hawkes on the same scenario.

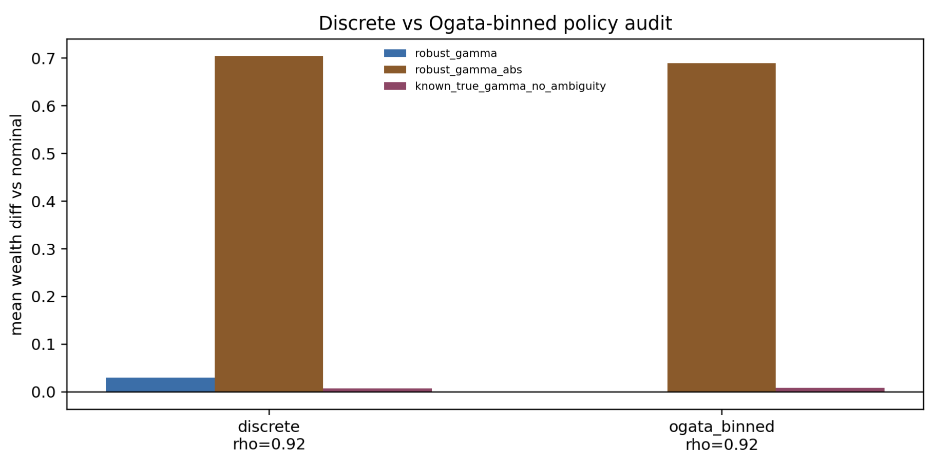

Figure 4: Focused policy audit under fast discrete arrivals and Ogata-binned event arrivals for $\hat{\rho} = 0.92$.

flat but positive relative to nominal, with mean-wealth difference 0.0125, while absolute-Gamma and the liquidity guard each improve mean wealth by 0.342. In this reduced run, fill rates are identical across policies at 0.943, the side-time no-quote fraction is 0.064, and the full no-quote time fraction is zero. The result is not a production queue simulator, but it provides an event-time execution stress test for the checked regime.

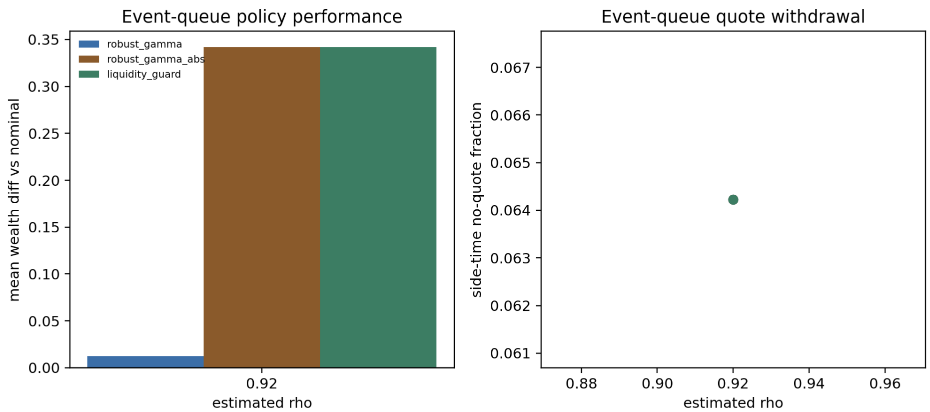

Figure 5: Reduced event-queue policy performance and shared side-time no-quote fraction.

6.6 Public LOBSTER Top-Of-Book Replay

As a public-data execution check, we replay the policy family on actual LOBSTER message/order-book streams. The replay uses the first 80,000 synchronized messages per ticker, posts active policies at the displayed L1 quotes subject to an inventory cap, depletes a simple queue-ahead proxy using observed execution sizes, and marks to the observed midprice. Under the calibrated side-aware spectral radii, all tested policies remain inside the L1 activity threshold, so the replay intentionally reports no artificial separation: mean terminal wealth is 341.691, mean fills are 3071.8, and side-time no-quote share is 0.198 for each policy. Under the near-critical stress setting $\hat{\rho} = 0.97$, nominal and relative-slack robust-Gamma remain active on the same real event streams, while absolute-Gamma and the liquidity guard fully withdraw: their mean wealth difference versus nominal is -341.691, mean fills are zero, and side-time no-quote share is one. This is not a production queue simulator, but it shows that the quote-withdrawal transition can be replayed on real event paths and L1 prices.

We also run an offset-aware L1 replay. It converts each policy's continuous offsets into visible quote states: join the displayed best quote, improve inside the spread, rest away from L1, or withdraw. Away quotes are not filled, because one-level data cannot identify hidden depth. Under calibrated side-aware spectral radii, tick rounding still makes policy PnL coincide, but the replay explains the quote placement: mean side-time is 0.024 joining, 0.428 improving, 0.448 away from L1, and 0.101 withdrawn. Under $\hat{\rho} = 0.97$, absolute-Gamma and the liquidity guard fully withdraw. This is a stronger public-data audit than the binary replay, while still stopping short of production execution claims.

Finally, we run a displayed-depth replay on the uniform level-10 LOBSTER panel. This removes the one-level limitation for away quotes: if a rounded quote price rests inside the displayed ten-level ladder, the replay initializes queue-ahead from all better displayed levels plus a same-level queue fraction, and observed executions and visible depth changes can deplete that queue. Under calibrated side-aware spectral radii, policies still coincide after tick rounding, but the state mix is more informative: mean side-time is 0.018 joining L1, 0.428 improving, 0.408 resting at visible depth, and 0.146 withdrawn, with mean visible depth rank 2.431 and mean fills 1420.2. Under $\hat{\rho} = 0.97$, absolute-Gamma and the liquidity guard fully withdraw, with mean wealth difference -120.216 versus nominal. The replay remains conservative about hidden liquidity and exact priority, but it is no longer L1-only.

As a stricter execution audit, we add a priority-aware level-10 replay. At each quote reset, the synthetic order is placed behind the full displayed size at the same price, later same-price limit orders are tracked by order identifier as being behind the synthetic quote, and cancellations/executions of those behind orders are not allowed to deplete our queue position. Queued fills are credited only from residual same-price execution volume after displayed queue ahead has been depleted; exact queue exhaustion with no residual volume is recorded but not filled, and partial fills are credited by residual lots. Under calibrated side-aware spectral radii, this stricter replay exposes how much of the displayed-depth result relied on a heuristic same-level queue cap: mean fill events are 21.6 but mean filled lots are only 15.53, mean terminal wealth is -7.839, and mean side-time is 0.019 joining L1, 0.530 improving, 0.422 resting at visible depth, and 0.029 withdrawn. Under $\hat{\rho} = 0.97$, absolute-Gamma and liquidity-guard policies fully withdraw and avoid the nominal priority-replay loss, giving mean wealth difference $+5.964$ versus nominal. The regenerated artifact also records queue-position invariants: in the calibrated baseline, mean initial displayed queue ahead is 62.03 lots across 1810.4 visible quote resets, queued fills average 1.2, inside-spread improve fills average 20.4, partial-fill events average 7.6, the maximum zero-residual prevented-fill count is 1, and the maximum queue-violation count is zero. This is the closest public-data replay in the package to queue-position accounting; it still cannot observe hidden liquidity or recover order identifiers for anonymous queue ahead present before the synthetic quote arrives.

As a depth supplement, we also replay the deepest public LOBSTER files available for the sample panel: level-50 AAPL and MSFT on their shorter one-hour public window. This supplement uses all 50 displayed levels in those files and records mean fill events 23.0, mean filled lots 16.82, mean queued fills 2.5, mean improve fills 20.5, mean partial-fill events 8.0, max visible depth rank 5, and maximum queue-violation count zero. Because only two tickers have public level-50 samples in the mirror, this is a supplement rather than the headline five-ticker panel.

We also run a priority-assumption sensitivity grid for the nominal policy. The grid varies the fraction of displayed same-price queue placed ahead of the synthetic order over $\{0, 0.5, 1\}$ and a queue-stress multiplier over $\{1, 1.5, 2\}$. The multiplier is applied to the public displayed queue-ahead proxy as a conservative stress test; it is not evidence that hidden depth has been observed. On the calibrated level-10 panel, the optimistic displayed priority setting $(0, 1)$ gives mean filled lots 16.53; the strict displayed-priority baseline $(1, 1)$ gives mean filled lots 15.53; and the most conservative tested queue-stress setting $(1, 2)$ gives mean filled lots 14.87. The small numerical range is a negative but useful empirical result: public displayed books rarely identify enough residual same-price execution volume to make hidden-priority assumptions strongly estimable. Across the sensitivity grid, the maximum queue-violation count remains zero.

As a final queue-audit layer, we reconstruct the observable level-10 book from messages and compare it to the official LOBSTER snapshots. The reconstructor starts from the first official snapshot, tracks all post-start order IDs, and re-anchors every 100 messages to measure local reconstruction quality rather than all-day truncation drift. Across 399,995 processed events and 40,000 compared snapshots, panel mean top-of-book price match is 0.9997, full 10-level price match is 0.9376, top-of-book size MAE is 0.056 shares, and full-depth size MAE is 38.1 shares. This supports local near-touch priority analysis from public messages while keeping deeper queue-position claims conservative.

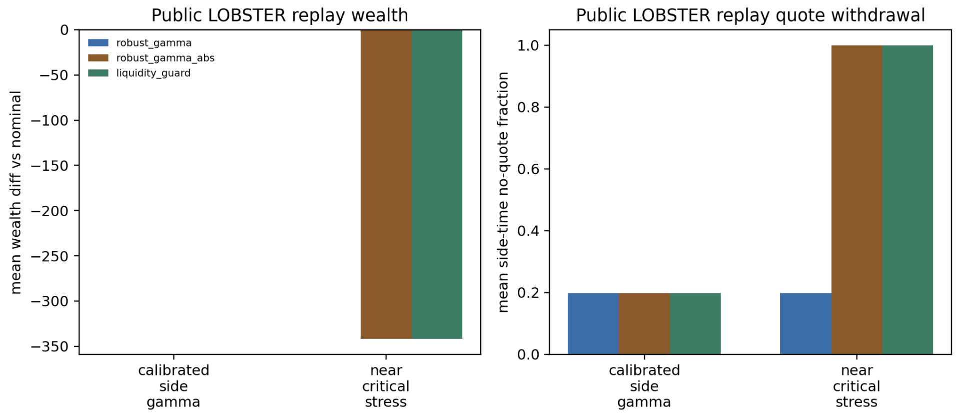

Figure 6: Public LOBSTER top-of-book replay on actual message/order-book streams. The binary top-of-book rule makes calibrated policies coincide, while near-critical absolute-Gamma ambiguity forces quote withdrawal on the same data paths.

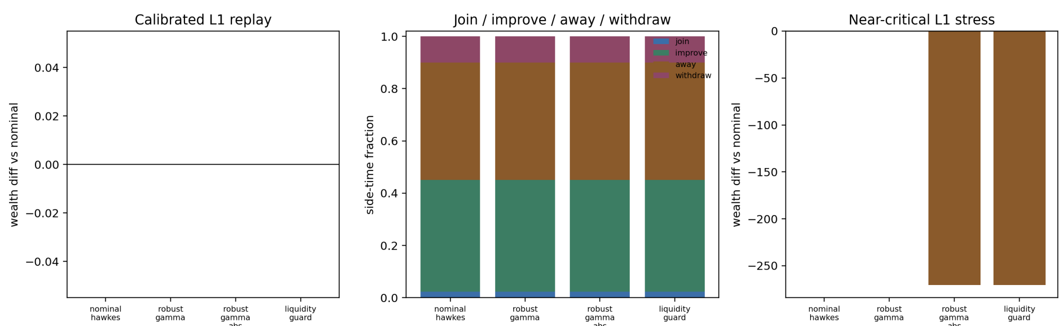

Figure 7: Offset-aware L1 quote replay. The calibrated public sample still does not separate policy wealth after tick rounding, but it reveals visible quote state fractions and shows that near-critical absolute-Gamma ambiguity produces full withdrawal.

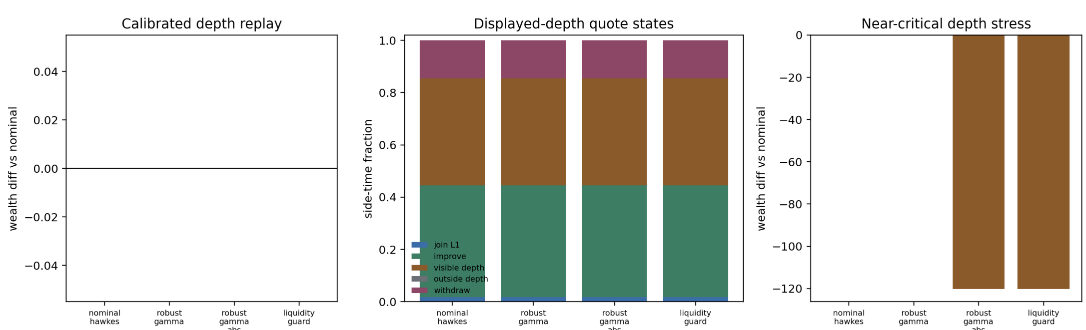

Figure 8: Displayed-depth level-10 LOBSTER quote replay. Away-from-L1 quotes can fill when they rest inside the visible book and observed executions/depth changes deplete displayed queue ahead.

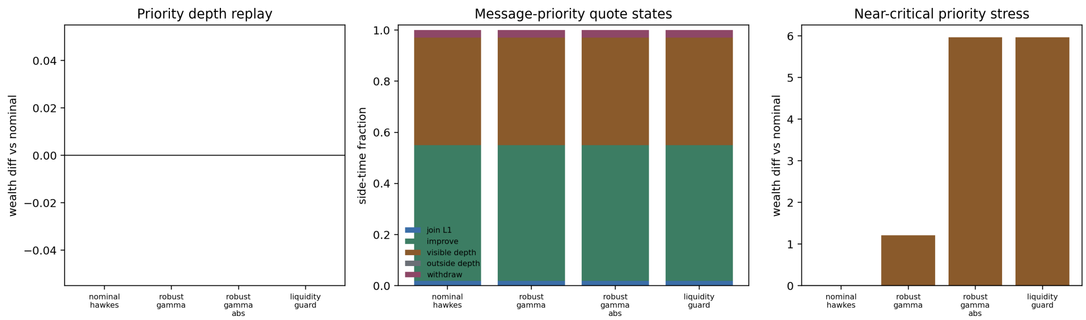

Figure 9: Residual-volume priority-aware level-10 LOBSTER quote replay. Same-price orders arriving after the synthetic quote are tracked behind it, queued fills require residual execution volume after displayed queue-ahead depletion, and partial fills are credited by residual lots.

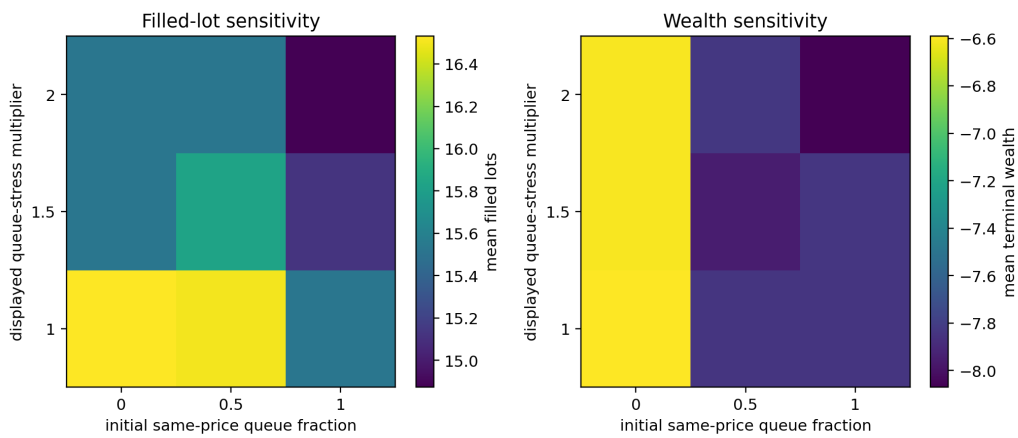

Figure 10: Priority-assumption sensitivity for the nominal level-10 replay. The grid varies initial same-price displayed priority and a displayed queue-stress multiplier; filled lots remain rare across the grid, exposing the public-data limit of same-price priority identification.

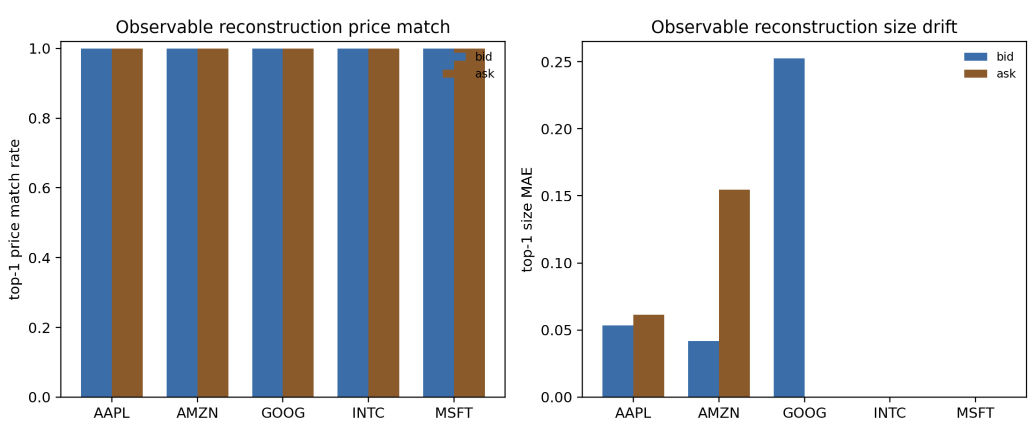

Figure 11: Observable order-book reconstruction audit. Re-anchored message-level reconstruction matches top-of-book prices almost exactly, while full-depth size drift remains visible.

6.7 Simulator Validation

The slow Ogata reference simulator exposes an important numerical issue. Coarse time discretization can create spurious critical amplification. At $\rho = 0.90$, the relative mean-count bias drops from 3.15 at $\Delta t = 0.08$ to 0.03 at $\Delta t = 0.01$. At $\rho = 0.97$, even $\Delta t = 0.005$ leaves about 0.15 relative bias in the current short-horizon setting.

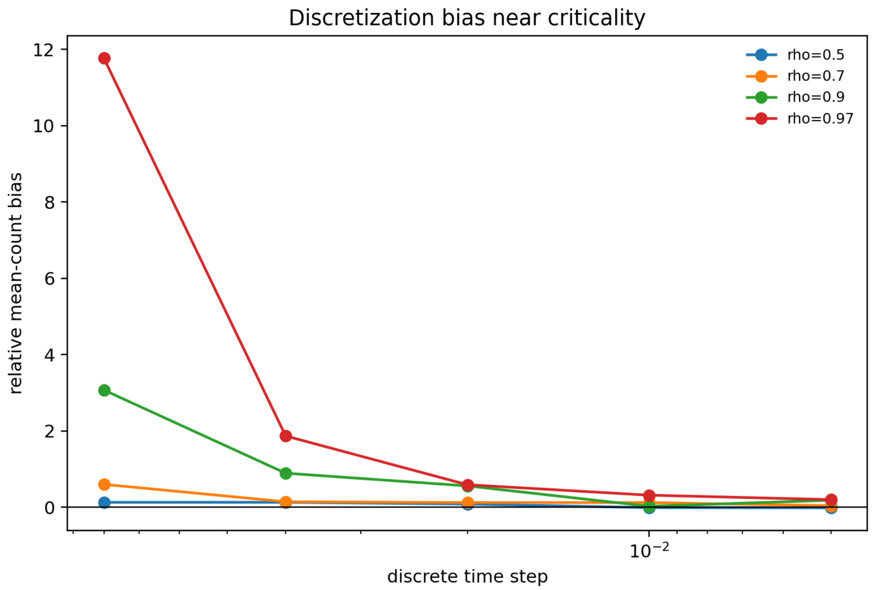

Figure 12: Time-discretization bias grows sharply near criticality.

6.8 Calibration Noise

A simple Fano-based criticality estimator confirms that near-critical calibration is fragile. For $\rho = 0.97$, short horizons under-estimate criticality, while longer horizons can over-shoot because the count variance is dominated by rare clusters. This motivates reporting uncertainty intervals for all empirical $\hat{\rho}$ estimates.

6.9 Robust DP

The finite-scenario DP now includes explicit bid-side and ask-side no-quote actions. In the full refreshed run, the raw DP action grid contains 9,947 two-sided quote states, 4,241 bid-only states, 4,241 ask-only states, and 579 full no-quote states. The first-step, zero-inventory policy switches to full no-quote for $\hat{\rho} \geq 0.97$ across the tested ambiguity radii. Averaged over the DP grid, no-quote rates are 45.7% under relative-slack ambiguity and 49.6% under absolute-Gamma ambiguity; full two-sided withdrawal rates are 2.9% and 3.2%, respectively. This is still a reduced Bellman approximation, not a queue-calibrated quote-refusal simulator comparable one-for-one with [5].

To connect the local optimizer theorem to the implementation, we also run a quote-map sensitivity diagnostic. For the smooth implemented half-spread, nominal and relative-Gamma robust policies recover critical exponent 3.000 for $d\delta/d\rho$, while absolute-Gamma robust policies approach exponent 4.000 in the uncapped interior region. Capped or no-quote points are marked as nonsmooth active-set cases rather than treated as classical derivatives.

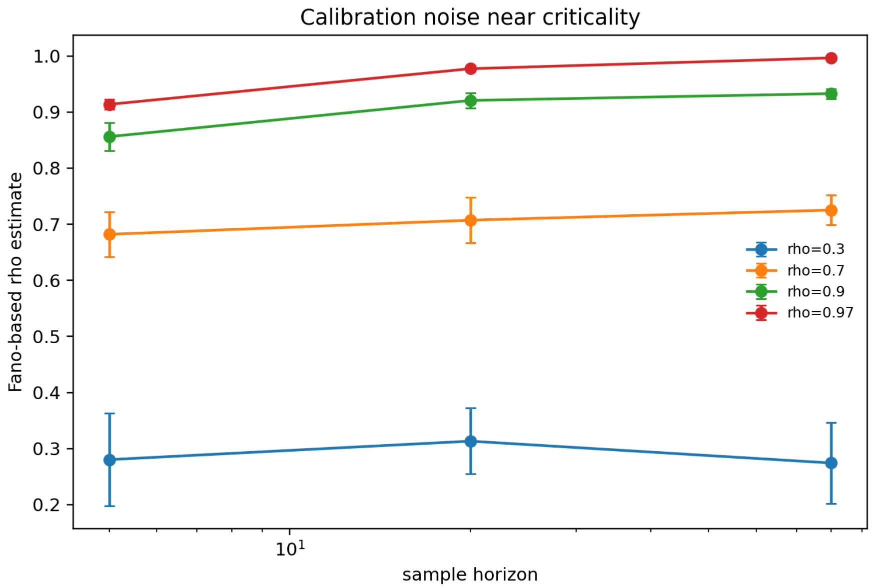

Figure 13: Fano-based criticality estimates across horizons.

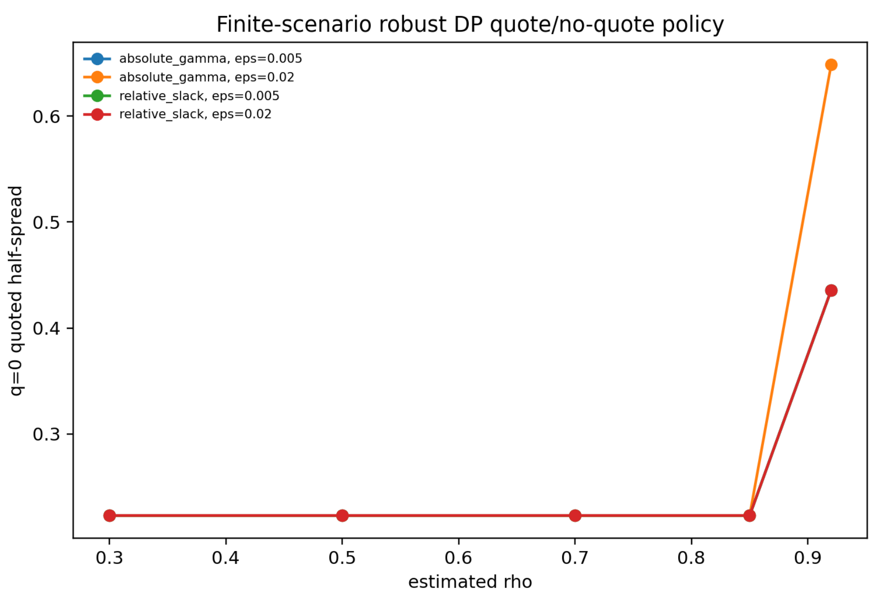

Figure 14: Reduced robust-DP quoted half-spread and no-quote switching as criticality increases.

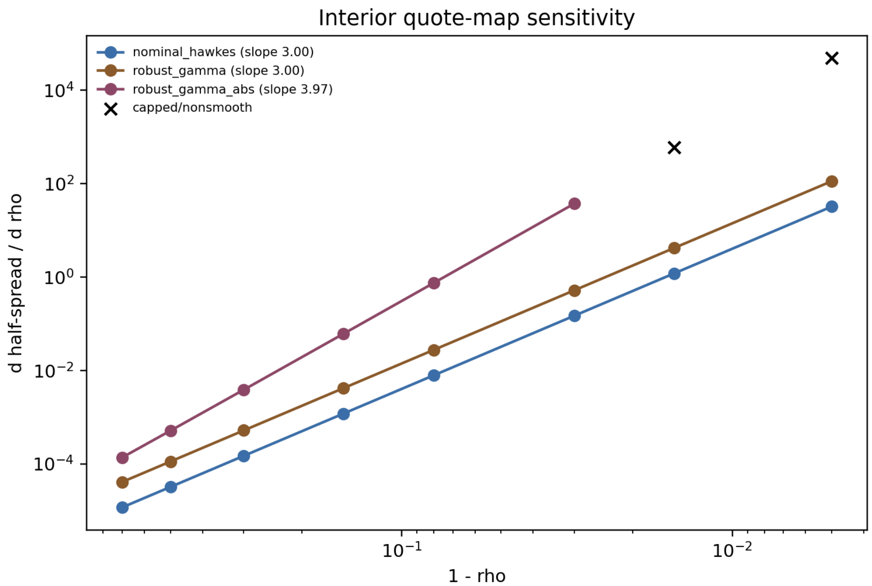

Figure 15: Interior quote-map sensitivity. Smooth relative-Gamma policies have third-order derivative blowup, while absolute-Gamma ambiguity has one additional critical derivative factor until spread caps bind.

6.10 Public Data Panels

The public datasets are not used to prove the structural ambiguity theorem, because the true Γ is unobserved. They establish empirical plausibility for clustered order flow and overdispersed liquidity states. Binance aggregate trades are fetched from the public archive at https://data.binance.vision/data/spot/daily/aggTrades/.

Ticker	Rows	Events/sec	Event Fano	Mean spread
AAPL	118497	5.064	19.460	0.155
AMZN	57515	2.458	13.329	0.136
GOOG	49482	2.115	17.185	0.311
INTC	404986	17.307	220.650	0.013
MSFT	411409	17.582	196.485	0.013

Table 4: Public LOBSTER sample sanity panel.

6.11 Corrected Hawkes Calibration Diagnostics

Before interpreting real-data Hawkes fits, we validated the corrected single-exponential MLE on Ogata-simulated paths with known parameters. The validation recovers moderate and near-critical branching ratios without beta-bound failures.

We additionally validate the fixed-beta marked multivariate estimator on three-type Ogata Hawkes paths with known branching matrices. For target spectral radii 0.35 and 0.65 at $\beta = 3$, the marked estimator has spectral-radius MAE 0.025 and 0.023, relative Frobenius branching-matrix error 0.234 and 0.133, and success rate one. This check is important because the public-data

Symbol	Agg. trades	Agg. trades/sec	Event Fano	ACF1
BTCUSDT	1364603	15.794	200.274	0.342
ETHUSDT	597417	6.915	29.730	0.241
SOLUSDT	218278	2.526	12.240	0.201

Table 5: Public Binance aggregate-trade event panel for 2024-01-15.

Symbol	Rows	Rel. spread bps	Return std	Depth Fano
BTC	10081	0.163	0.000687	33.989
ETH	10081	0.591	0.000910	513.703
SOL	10080	0.911	0.001110	1697.605

Table 6: Public crypto L2 depth sanity panel.

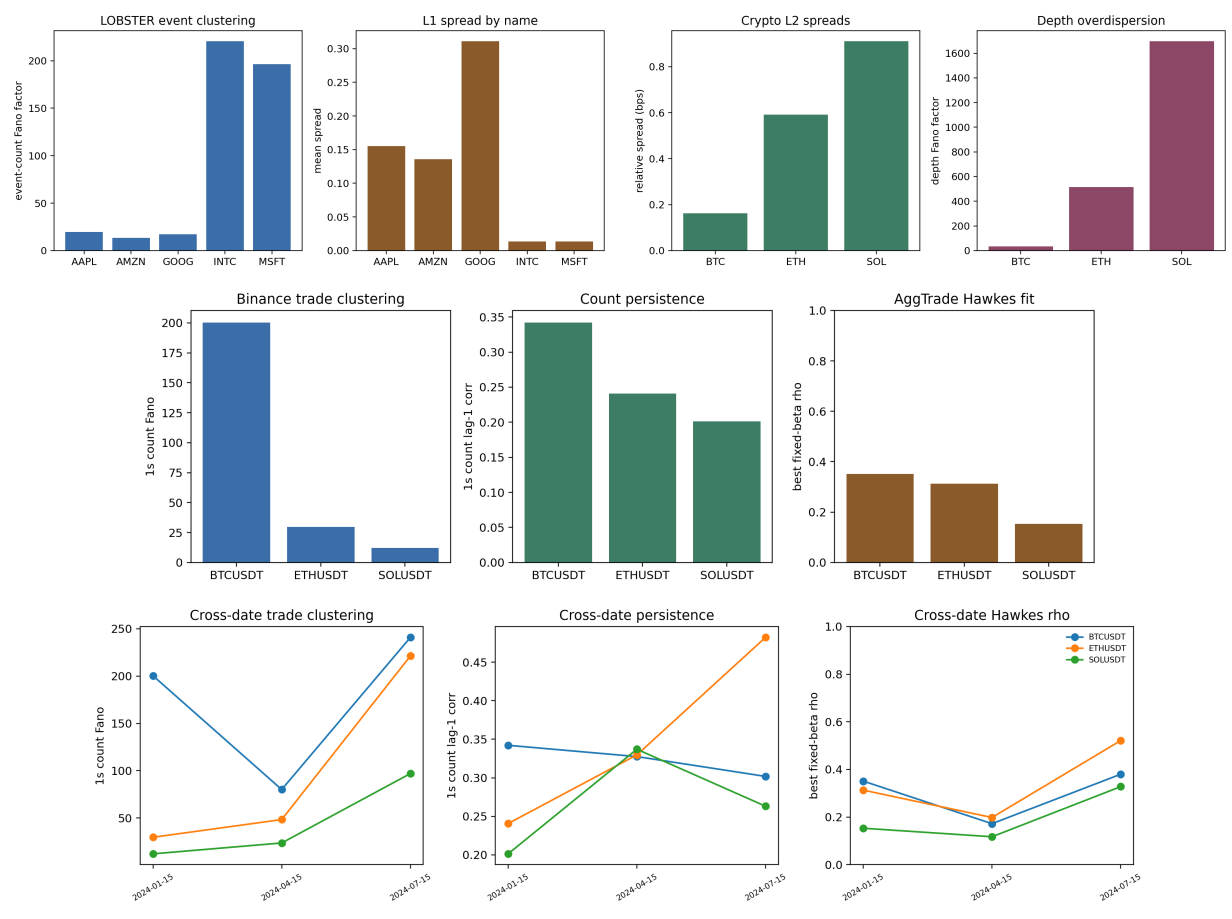

Figure 16: Public-data sanity figures: LOBSTER event clustering and spreads, crypto L2 spreads and depth overdispersion, Binance aggregate-trade clustering, fixed-beta Hawkes diagnostics, and cross-date robustness.

ρ true	β true	ρ MAE	β MAPE	Beta cap rate
0.30	2.0	0.043	0.201	0.000
0.70	4.0	0.019	0.026	0.000
0.90	8.0	0.016	0.039	0.000

Table 7: Synthetic Ogata validation for the corrected Hawkes MLE.

marked calibrations below are only meaningful if the estimator recovers a known multitype Γ under controlled simulation.

ρ true	Mean events	ρ MAE	$\|\hat{\Gamma} - \Gamma\|_F / \|\Gamma\|_F$	Success
0.35	1209.8	0.025	0.234	1.000
0.65	2171.2	0.023	0.133	1.000

Table 8: Synthetic Ogata validation for the fixed-beta marked multivariate Hawkes estimator.

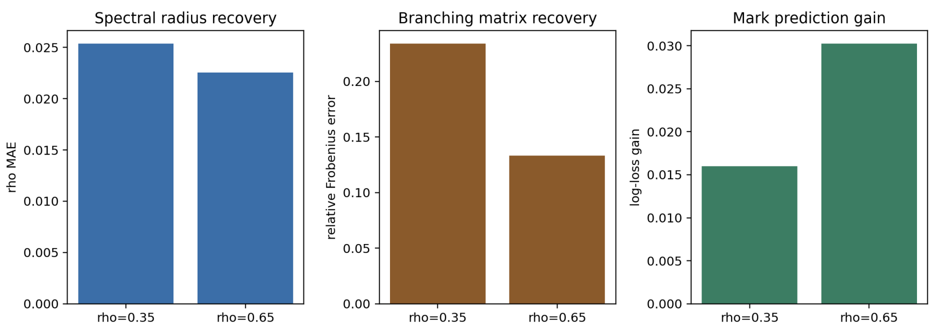

Figure 17: Marked multivariate Ogata validation: spectral-radius recovery, branching-matrix recovery, and conditional mark-prediction gain.

The Binance aggregate-trade panel provides a separate public crypto event check. On 2024-01-15, BTCUSDT, ETHUSDT, and SOLUSDT have 218k to 1.36m aggregate trades, one-second event-count Fano factors from 12.24 to 200.27, and lag-one count autocorrelations from 0.201 to 0.342. Fixed beta Hawkes fits over all aggregate trades, buy-aggressor trades, and sell-aggressor trades select $\beta = 100$ in the current grid. All-trade branching ratios are 0.351 for BTCUSDT, 0.313 for ETHUSDT, and 0.153 for SOLUSDT; sell-aggressor branching is higher for BTCUSDT and ETHUSDT, at 0.493 and 0.463. This is useful cross-asset evidence of clustered event flow, but aggregate trades do not include order placement, cancellation, or queue state, so the panel is not an execution benchmark.

To avoid relying on one crypto day, we repeat the Binance check on 2024-01-15, 2024-04-15, and 2024-07-15. Across all-event fixed-beta fits, branching-ratio ranges are 0.172–0.380 for BTCUSDT, 0.198–0.521 for ETHUSDT, and 0.117–0.328 for SOLUSDT. The median all-event branching ratios remain 0.351, 0.313, and 0.153, respectively.

On raw LOBSTER message arrivals, however, the all-message single-exponential fit remains a misspecification diagnostic rather than a structural calibration target. The corrected all-message panel still places β at the upper bound for every ticker under the 100k-event cap, with fitted branching ratios between 0.64 and 0.76. Event-type splits and fixed-beta profiles show that branching estimates are sensitive to the assumed decay scale; timestamp aggregation sharply reduces the fitted decay, with median β falling from 148.4 on raw events to 23.4 at 1ms aggregation and 1.46 at 10ms aggregation. A fixed-beta two-scale diagnostic improves the empirical picture but does not remove the misspecification warning: among the tested pairs, every ticker/event-group selects $(\beta_{\text{slow}}, \beta_{\text{fast}}) = (1, 100)$, and median fast-excitation shares are 0.764 for all messages, 0.723 for limits, 0.660 for cancel/delete events, and 0.871 for executions. Execution events remain the hardest to fit, with median residual KS statistic about 0.415. This supports the methodological claim that raw unmarked message flows mix multiple clocks and packet-level bursts, so robust market-making experiments should use marked or multiscale calibration diagnostics, or report residual diagnostics [1, 18].

The current package therefore adds a grouped marked multivariate Hawkes fit on limit, cancel/delete, and execution event groups. With fixed exponential decay, AIC selects $\beta = 100$ for all five public LOBSTER tickers. The estimated branching matrices are stable, with spectral radii from 0.367 to 0.622, and the conditional mark log-loss improves over unconditional event-frequency prediction by 0.005 to 0.114 nats per event. The median branching matrix has the strongest source column from execution events, with median total outgoing branching about 0.923, and the largest median target row into limit intensity, about 0.897. Residual KS statistics still range from 0.050 to 0.217, so the fit should be read as a reproducible marked calibration diagnostic, not as a production-grade structural Γ estimate.

Finally, to align the empirical object with the bid/ask control model, we fit a six-mark side-aware Hawkes model using LOBSTER direction labels: limit, cancel/delete, and execution events on buy and sell sides. AIC again selects $\beta = 100$ for all five tickers. The full six-mark branching matrices remain stable, with spectral radii from 0.369 to 0.624, while conditional six-mark log-loss improves over unconditional mark frequencies by 0.101 to 0.292 nats per event. Median side-aggregated branching is strongest within side, about 1.354 from buy to buy and 1.403 from sell to sell, with smaller cross-side totals of about 0.492 from buy to sell and 0.414 from sell to buy. The aggregate row sums can exceed one because they sum many target/source entries; stability is assessed by the spectral radius of the full six-mark matrix.

We then condition the six-mark time-rescaling residuals on observable L1 state variables: event group, side, within-ticker size bucket, spread bucket, top-depth bucket, and order-book imbalance bucket. This is intentionally a misspecification audit rather than another fitted model. Across the five public LOBSTER samples, wide-spread states have median residual mean 0.886 and median KS statistic 0.220, while tight-spread states have median residual mean 1.110 and median KS statistic 0.092. Large-size events have residual mean 0.828 versus 1.018 for small events. Execution marks show the largest conditional mark-prediction gain, with median log-loss improvement 0.384 nats per event. The result sharpens the empirical conclusion: side-aware excitation is informative, but the residuals still depend materially on size and book state.

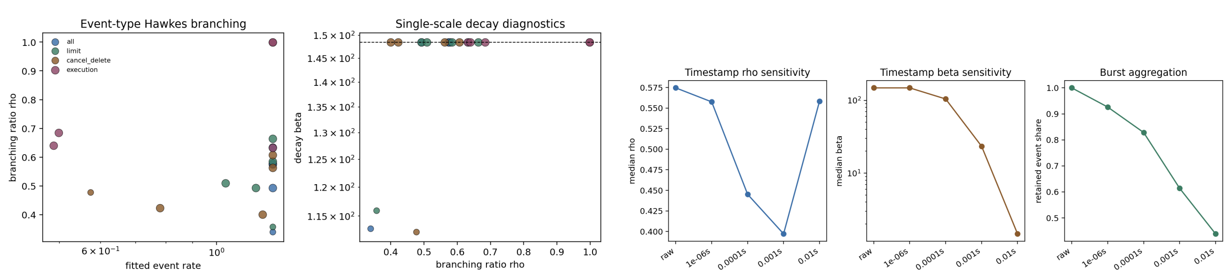

Figure 18: Corrected public-data Hawkes diagnostics: event-type branching/decay and timestamp-resolution sensitivity.

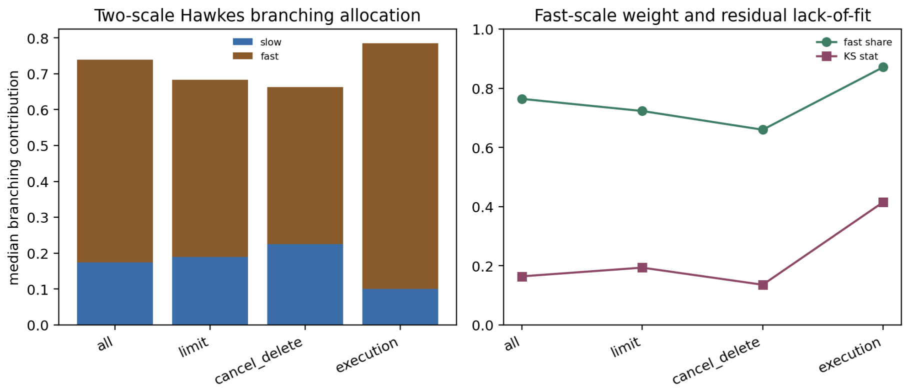

Figure 19: Fixed-beta two-scale public-data Hawkes diagnostics. The selected two-scale fits concentrate excitation on the fast component, especially for execution events, while residual tests still flag misspecification.

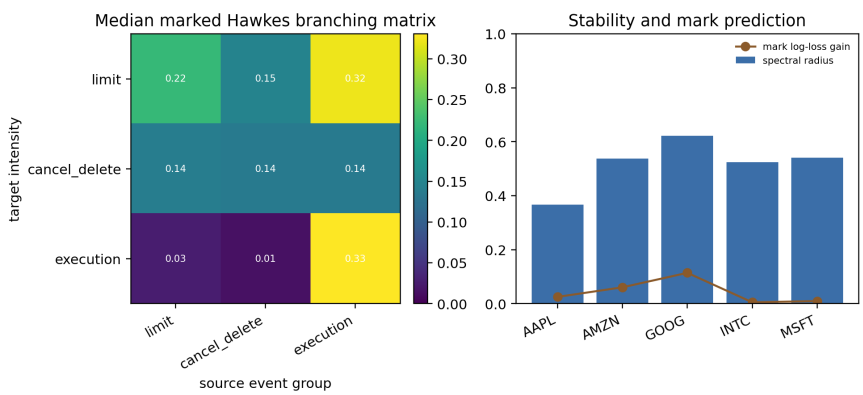

Figure 20: Grouped marked multivariate Hawkes diagnostics on LOBSTER limit, cancel/delete, and execution events. The fixed-beta fits estimate stable branching matrices and improve conditional mark prediction, but residual tests still motivate richer side, size, and multiscale calibration.

7 Scope Conditions And Extensions

The current package gives a conditional multitype HJBI control bridge for the stated exponential Hawkes model class. The proof layers are compact truncated HJBI verification under compact-PDMP DPP/comparison hypotheses, untruncated Lyapunov stability with $p > 1$ tail control, pairwise weighted comparison and uniqueness in fixed-boundary W_v-weighted subclasses, local differentiability of smooth interior quote optimizers, and an assembled end-to-end theorem combining these layers. It does not claim the same theorem for every possible state dependent, learned, multiscale, or power law Hawkes kernel without rechecking the stated Lyapunov and comparison assumptions. It also treats no-quote and active adversary switches as nonsmooth control boundaries rather than classical optimizer derivatives.

The current top-of-book, L1, displayed-depth, residual-volume priority-aware level-10, deepest-public level-50, priority-sensitivity, and observable-reconstruction LOBSTER audits should still be treated as public-data execution audits rather than production execution claims, because hidden

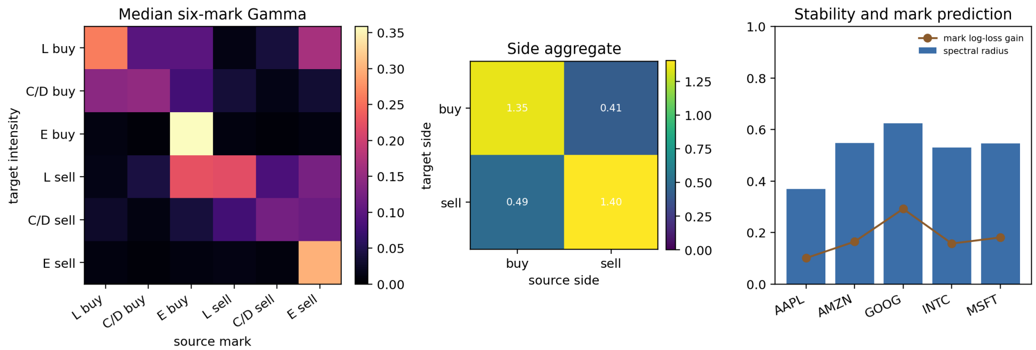

Figure 21: Side-aware marked multivariate Hawkes diagnostics using LOBSTER direction labels. The six-mark fits bridge the empirical calibration object toward bid/ask robust control, while remaining fixed-decay and message-side rather than queue-position or size-aware.

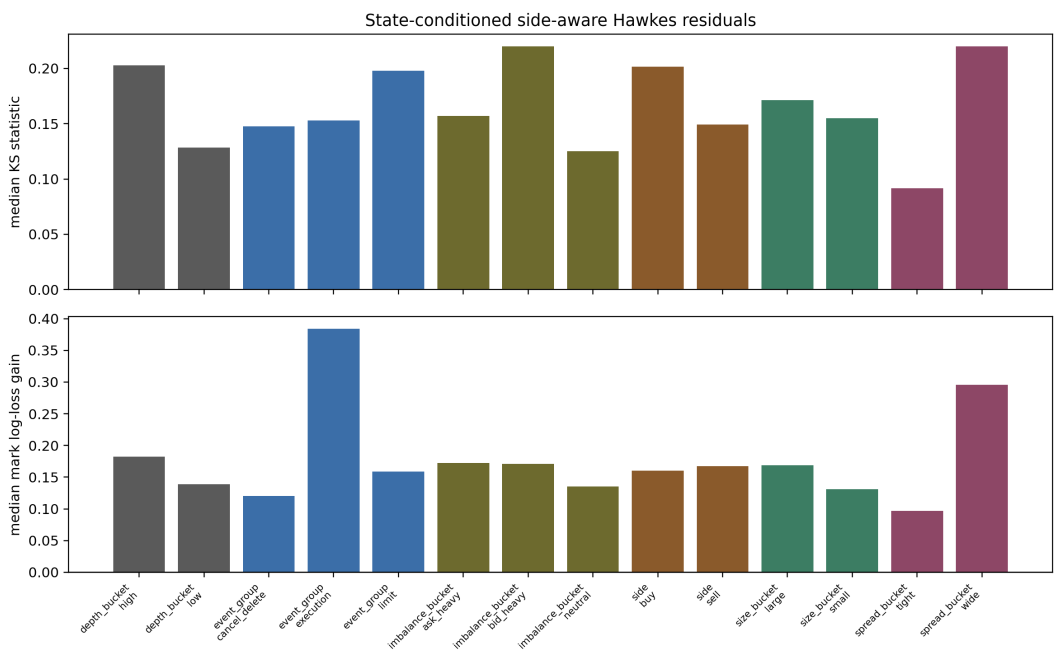

Figure 22: State-conditioned residual diagnostics for the side-aware six-mark Hawkes fit. Residual deviations and mark-prediction gains vary by spread, depth, imbalance, and event-size bucket, so the public-data calibration should be treated as a diagnostic input rather than a complete structural law.

liquidity and anonymous initial queue priority remain unobserved. Binance aggregate trades should remain event-clustering evidence rather than a queue-calibration source. These scope conditions discipline the paper around its correct novelty: structural Hawkes ambiguity, criticality amplification, and calibration fragility in Hawkes market-making experiments.

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